22/07/2026

CODE-A

Aakash
Medical | IIT-JEE | Foundations

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FORTNIGHTLY TEST FOR NEET-2027 (XII-PASSED)

MM: 720 **FT-03A** **Time: 180 Mins.**

Answers

1.	(3)	37.	(3)	73.	(1)	109.	(2)	145.	(4)
2.	(3)	38.	(3)	74.	(2)	110.	(3)	146.	(1)
3.	(4)	39.	(2)	75.	(1)	111.	(2)	147.	(2)
4.	(3)	40.	(2)	76.	(3)	112.	(1)	148.	(1)
5.	(4)	41.	(2)	77.	(3)	113.	(1)	149.	(2)
6.	(2)	42.	(2)	78.	(3)	114.	(2)	150.	(1)
7.	(3)	43.	(4)	79.	(4)	115.	(1)	151.	(1)
8.	(2)	44.	(3)	80.	(4)	116.	(2)	152.	(2)
9.	(3)	45.	(2)	81.	(3)	117.	(4)	153.	(2)
10.	(3)	46.	(3)	82.	(2)	118.	(3)	154.	(2)
11.	(4)	47.	(1)	83.	(3)	119.	(3)	155.	(3)
12.	(1)	48.	(2)	84.	(2)	120.	(1)	156.	(4)
13.	(1)	49.	(2)	85.	(1)	121.	(2)	157.	(1)
14.	(3)	50.	(3)	86.	(2)	122.	(2)	158.	(2)
15.	(4)	51.	(2)	87.	(2)	123.	(1)	159.	(3)
16.	(3)	52.	(3)	88.	(3)	124.	(3)	160.	(3)
17.	(1)	53.	(3)	89.	(2)	125.	(2)	161.	(1)
18.	(4)	54.	(1)	90.	(4)	126.	(1)	162.	(2)
19.	(2)	55.	(2)	91.	(3)	127.	(4)	163.	(1)
20.	(2)	56.	(4)	92.	(4)	128.	(2)	164.	(3)
21.	(2)	57.	(2)	93.	(1)	129.	(4)	165.	(2)
22.	(4)	58.	(3)	94.	(3)	130.	(3)	166.	(2)
23.	(2)	59.	(4)	95.	(4)	131.	(3)	167.	(3)
24.	(3)	60.	(4)	96.	(3)	132.	(1)	168.	(3)
25.	(1)	61.	(1)	97.	(4)	133.	(3)	169.	(2)
26.	(3)	62.	(3)	98.	(2)	134.	(3)	170.	(3)
27.	(1)	63.	(1)	99.	(2)	135.	(2)	171.	(2)
28.	(4)	64.	(4)	100.	(1)	136.	(2)	172.	(2)
29.	(1)	65.	(4)	101.	(3)	137.	(3)	173.	(3)
30.	(1)	66.	(4)	102.	(2)	138.	(4)	174.	(1)
31.	(4)	67.	(4)	103.	(3)	139.	(3)	175.	(2)
32.	(3)	68.	(4)	104.	(3)	140.	(4)	176.	(2)
33.	(2)	69.	(3)	105.	(3)	141.	(2)	177.	(3)
34.	(2)	70.	(2)	106.	(3)	142.	(4)	178.	(4)
35.	(3)	71.	(2)	107.	(2)	143.	(2)	179.	(2)
36.	(2)	72.	(2)	108.	(4)	144.	(3)	180.	(1)

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Hints and Solutions

PHYSICS

(1) **Answer :** (3)
Solution:

Work done by gravity = $m\vec{g}.\vec{d} = mgd\cos 90^\circ = \text{zero}$

(2) **Answer :** (3)
Solution:

W = $\vec{F}\ .\ \Delta\vec{r} = \vec{F}.\left(\vec{r}_2 - \vec{r}_1\right)$

= $\left(\hat{i} + \hat{j}\right).\ \left(2\hat{i} + 3\hat{j} - 3\hat{i} + 4\hat{j}\right)$

= $\left(\hat{i} + \hat{j}\right).\left(-\hat{i} + 7\hat{j}\right)$

= – 1 + 7 = + 6J

(3) **Answer :** (4)
Solution:

$\because\ \ x = \frac{t}{3}$

$v = \frac{dx}{dt} = \frac{1}{3}$

$a = \frac{dv}{dt} = 0$

$\therefore W = F{\cdot}x$

= 0

(4) **Answer :** (3)
Solution:

Frictional force can do positive, negative or zero work. Depending on the work done by friction, kinetic energy may increase or decrease.

(5) **Answer :** (4)
Solution:

$$W = \int_1^3 F\,dx = \int_1^3 (3x^2 - 2x + 5)\,dx$$

$$= \left[x^3 - x^2 + 5x\right]_1^3 = (27 - 9 + 15) - (1 - 1 + 5) = 28\ \text{J}$$

(6) **Answer :** (2)
Solution:

$a_{\text{cm}} = \frac{m_1a_1 + m_2a_2}{m_1 + m_2} = \frac{m\times 0 + m\times a}{2m}$

$a_{\text{cm}} = \frac{a}{2}$

(7) **Answer :** (3)
Solution:

In a couple, total external force is zero because both forces are equal in magnitude but opposite in direction. The total torque on a system is independent of the origin if the total external force is zero.

(8) **Answer :** (2)
Hint:

$\alpha = \frac{\omega_f - \omega_i}{t}$ & $\theta = \omega_0 t + \frac{1}{2}\alpha t^2$

Solution:

f_0 = 420 rev/min

$\omega_0 = 2\pi \times 7 = 14\pi$ rad/s

At t = 10 s, $\omega = 0$

$\alpha = -\frac{14\pi}{10} = -\frac{7\pi}{5}$ rad/s^2

$\theta = \omega_0 t + \frac{1}{2}\alpha t^2$

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$= 14\pi \times 10^{-\frac{1}{2}} \times \frac{7\pi}{5} \times 100$

θ = 70π rad

Number of revolutions = $\frac{\theta}{2\pi}$

$= \frac{70\pi}{2\pi} = 35$ revolutions

(9) **Answer :** (3)
Solution:

$\vec{v} = \vec{\omega} \times \vec{r}$

$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 1 \\ 1 & 2 & 3 \end{vmatrix}$

$= \hat{i}(-3-2) - \hat{j}(3-1) + \hat{k}(2+1)$

$= \left(-5\hat{i} - 2\hat{j} + 3\hat{k}\right)$ m/s

(10) **Answer :** (3)
Solution:

Angular momentum $\vec{L} = \vec{r} \times \vec{p} =$ constant

$\therefore$ Torque $\vec{\tau}$ = zero and $\vec{\alpha}$ = zero

(11) **Answer :** (4)
Hint:
Potential energy is defined for conservative forces only.
Solution:
Negative work done by conservative forces increases PE of a system
$\Delta U = -W_{cons}$

(12) **Answer :** (1)
Solution:
Tension in the wire will be maximum at the lowest point while it will be minimum at the highest point. Hence, the wire will be least likely to break at highest point.

(13) **Answer :** (1)
Solution:

$F = -\frac{dU}{dx}$

$F = -\frac{d}{dx}\left(x^2 - 4x + 2\right)$

$= -2x + 4$

For equilibrium, $F = 0$

$-2x + 4 = 0$

$\boxed{x = 2\ m}$

(14) **Answer :** (3)
Solution:

$mk^2 = \frac{ml^2}{12}$

$k = \frac{l}{2\sqrt{3}}$

(15) **Answer :** (4)
Solution:
Centre of mass coincides with centre of gravity in the case where the gravitational field is uniform.

(16) **Answer :** (3)
Solution:
When dancer fold his hand, moment of inertia decreases by conservation of angular momentum and angular velocity increases. Therefore kinetic energy increases.

(17) **Answer :** (1)
Solution:

$h_2 = e^{(2n)}h_0$

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$= (0.5)^4 h_0$

$= 0.625$ m

(18) **Answer :** (4)

Solution:

$\Delta k = \frac{1}{2}\left(\frac{m_1 m_2}{m_1+m_2}\right)(V_1 - V_2)^2$

$= \frac{1}{2} \times \left[\frac{40\times 60}{40+60}\right]\left[4-2\right]^2 = 48$ J

(19) **Answer :** (2)

Hint:

Power required $P = \frac{W+KE}{\Delta t}$

Solution:

$P_r = \frac{mgh}{\Delta t} + \frac{\Delta KE}{\Delta t}$

$P_r = 3 \times 10 \times 10 + \frac{1}{2}\frac{mv^2}{t}$

$P_r = 300 + \frac{1}{2} \times 3 \times 100$

$P_r = 300 + 150$

$P_r = 450$ W

$\Rightarrow$ Power of motor $P_m = \frac{P_r}{\eta} = \frac{450}{0.6} = 750$ w

(20) **Answer :** (2)

Solution:

$P = \frac{dk}{dt}$ $\quad\quad\quad\quad$ $\therefore dk = Pdt$

$\Delta k = \int P dt = \int_2^4 \left(3t^2 - 2t + 1\right) dt$

$= 3\left[\frac{t^3}{3}\right]_2^4 - 2\left[\frac{t^2}{2}\right]_2^4 + [t]_2^4$

$= [4^3 - 2^3] - [4^2 - 2^2] + [4 - 2]$

$= [64 - 8] - [16 - 4] + 2$

$= 56 - 12 + 2 = 46$ J

(21) **Answer :** (2)

Hint:

Set of equal and opposite force acting on a body on different lines of action is called couple.

Solution:

$\vec{F}_1 + \vec{F}_2 = 0$

$\vec{F}_1 \times \vec{r}_1 + \vec{F}_2 \times \vec{r}_2 \neq 0$

(22) **Answer :** (4)

Solution:

$I = MR^2 + MR^2 = 2MR^2$

(23) **Answer :** (2)

Solution:

By conservation of angular momentum, $I_1\omega_1 = I_2\omega_2$

$\frac{2}{5}MR^2 \times \frac{2\pi}{T_1} = \frac{2}{5}M\left(\frac{R}{\sqrt{3}}\right)^2 \times \frac{2\pi}{T_2}$

$\Rightarrow T_2 = \frac{T_1}{3} = \frac{24}{3} = 8$ hours

(24) **Answer :** (3)

Hint:

Use, $\vec{v}_{com} = \frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1+m_2}$

Solution:

$\vec{v}_{com} = \frac{4\left(\hat{i}+\hat{j}-\hat{k}\right)+8\left(2\hat{i}-2\hat{j}+4\hat{k}\right)}{4+8}$

$= \frac{20\hat{i}-12\hat{j}+28\hat{k}}{12}$

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$= \frac{5}{3}\hat{i} - \hat{j} + \frac{7}{3}\hat{k}$

(25) Answer : (1)
Solution:
If $\vec{C} = \vec{A} \times \vec{B}$ then $\vec{C} \perp \vec{A}$ and $\vec{C} \perp \vec{B}$

(26) Answer : (3)
Hint:
Man and boat are the part of a system and initially centre of mass is at rest.
Solution:
$\vec{v}_{cm} = 0$

$m_1\overrightarrow{\Delta r_1} + m_2\overrightarrow{\Delta r_2} = 0$

Let displacement of boat is $\overrightarrow{\Delta r_2}$ w.r.t. ground

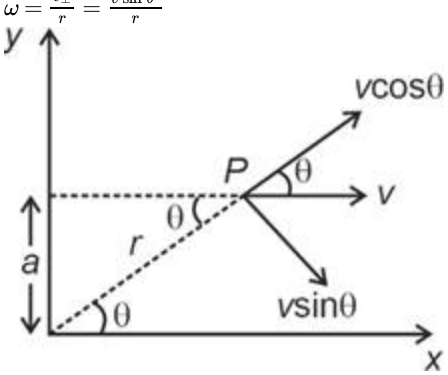
$100 \times \left(9 + \overrightarrow{\Delta r_2}\right) + 500\overrightarrow{\Delta r_2} = 0$

$900 + 600\overrightarrow{\Delta r_2} = 0$

$\overrightarrow{\Delta r_2} = \frac{-900}{600} = -1.5$ m

i.e. Boat moves 1.5 m relative to shore in the direction opposite to displacement of man.

(27) Answer : (1)
Hint:
Angular velocity $\omega = \frac{v_\perp}{r}$
Solution:
$\omega = \frac{v_\perp}{r} = \frac{v\sin\theta}{r}$



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$\frac{a}{r} = \sin\theta$

$r = \frac{a}{\sin\theta}$

$\omega = \frac{v\sin^2\theta}{a} = 2 \times \frac{1}{4\times 2} = \frac{1}{4}$ rad/s

(28) Answer : (4)
Solution:
Torque acting on a body is always equal to

$\vec{\tau} = \frac{d\vec{L}}{dt}$

As $\vec{L} = I\vec{\omega}$

$\vec{\tau} = \frac{d}{dt}(I\vec{\omega})$

$\vec{\tau} = I.\frac{d\vec{\omega}}{dt} + \frac{\vec{\omega}\,dI}{dt}$

· If ω is constant and *I* is varying, then

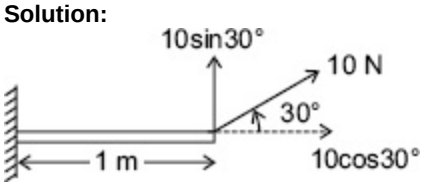
$\vec{\tau} = \vec{\omega}\frac{dI}{dt}$

· If *I* is constant and ω is varying, then

$\vec{\tau} = I\frac{d\vec{\omega}}{dt} = I\vec{\alpha}$

(29) Answer : (1)

Hint:
$\vec{\tau} = \vec{r} \times F$
Solution:



Torque acting on rod
τ = (10sin 30°) × 1 = 5 N × m

(30) **Answer :** (1)
Solution:

	MOI depends on
1.	Mass of body
2.	Mass distribution of body
3.	Position of axis of rotation
	MOI does not depend on
1.	Angular velocity
2.	Angular acceleration
3.	Torque
4.	Angular momentum

(31) **Answer :** (4)
Solution:
For perfectly elastic collision, $e = 1$
For inelastic collision, $0 < e < 1$
For perfectly inelastic collision, $e = 0$
In explosion, final kinetic energy is greater than initial kinetic energy.

(32) **Answer :** (3)
Solution:
$F = -\frac{dU}{dx} = -\left[\frac{d}{dx}\left(ax^2 - bx\right)\right] = -[2ax - b]$
$= -2ax + b$

(33) **Answer :** (2)
Solution:
By conservation of M.E
$\frac{1}{2}mV^2 = mgl$
$V = \sqrt{2gl}$

(34) **Answer :** (2)
Hint:
$\Delta U = -W_C$.
Solution:
$\Delta U = -\left(-3\hat{i}\right) \cdot \left(6\hat{i}\right)$
$\Delta U = 18$ J

(35) **Answer :** (3)
Hint:
$J = L_f - L_i = I\left(\omega_f - \omega_i\right) = mr^2\left(\omega_f - \omega_i\right)$
Solution:
$J = 3 \times \left(\frac{25}{100}\right)^2 \times (16) = 3 \times \left(\frac{1}{4}\right)^2 \times 16 = 3$ N m s

(36) **Answer :** (2)
Solution:
$I_1\omega_1 = I_2\omega_2$

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$\Rightarrow \quad \omega_2 = \frac{I\omega}{\left(\frac{5}{4}\right)I} = \frac{4}{5}\omega$

% change $\frac{\left(-\frac{\omega}{5}\right)\times 100}{\omega} = -20\%$

(37) **Answer :** (3)
Solution:
Work done by frictional force can be positive, negative or zero

(38) **Answer :** (3)
Solution:
$\overrightarrow{P_1} + \overrightarrow{P_2} + \overrightarrow{P_3} = 0$
$\overrightarrow{P_1} = -\left(\overrightarrow{P_2} + \overrightarrow{P_3}\right)$
$= -\left(2\hat{i} + 2\sqrt{3}\hat{j}\right)$
$\left|\overrightarrow{P_1}\right| = 2\sqrt{1 + \left(\sqrt{3}\right)^2}$
$= 2\sqrt{1+3}$ = 4 Ns
$KE = \frac{P_1^2}{2m_1} = \frac{4\times 4}{2\times 4} = 2$ J

(39) **Answer :** (2)
Solution:
$P = \frac{mgh}{t} = \frac{1\times 10\times 1}{10} = 1$ W

(40) **Answer :** (2)
Solution:
P = F.v
$P = Mv\frac{dV}{dt}$
$\int_0^V v dv = \int \frac{P}{m} dt$
$\frac{v^2}{2} = \frac{P}{m}t$
$v = \sqrt{\frac{2p}{m}}\, t^{\frac{1}{2}}$
$\frac{dx}{dt} = \sqrt{\frac{2p}{m}}\, t^{\frac{1}{2}}$
$\int dx = \int \sqrt{\frac{2p}{m}}\, t^{\frac{1}{2}} dt$
$x \propto t^{\frac{3}{2}}$

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(41) **Answer :** (2)
Solution:
At lower most position
$\frac{v^2}{l} = 5g$
T = 6 *mg*
At upper most position
T = 0
$\frac{v^2}{l} = g$
At horizontal position
T = 3 *mg*
Radial acceleration at uppermost position *mg* = *ma* ⇒ *a* = *g*
Radial acceleration at lowermost position *T* – *mg* = *ma*
6*mg* – *mg* = *ma*
a = 5*g*

(42) **Answer :** (2)
Solution:
Work done by friction = Loss of potential energy

= – 1 × 10 × 1.2 = – 12 J

(43) **Answer :** (4)
Solution:
The direction of angular velocity vector is along axis of rotation.

(44) **Answer :** (3)
Solution:
CGS unit of work is erg.

(45) **Answer :** (2)
Hint:
Because wall is in (y – z) plane, only x-component of velocity will change.
Solution:

$e = \frac{\text{velocity of separation}}{\text{velocity of approach}}$

$\frac{1}{2} = \frac{0-v_1}{2-0}$

$\overrightarrow{v}_1 = -\hat{i}$

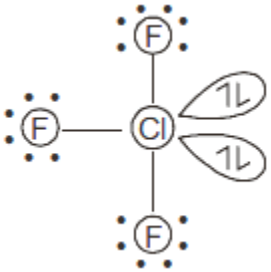
So, velocity just after collision $= -\hat{i} + 2\hat{j}$

CHEMISTRY

(46) **Answer :** (3)
Solution:
BF_3 is trigonal planar in shape and has 120° bond angle.

(47) **Answer :** (1)
Solution:
For 2nd group hydrides, on moving down the group metallic character of metals increases so ionic character of metal hydride increases. Hence the option (A) should be correct option.

(48) **Answer :** (2)
Solution:
The structure of ClF_3 is



The number of lone pair of electrons on central Cl is 2.

(49) **Answer :** (2)
Solution:
O = C = O carbon and oxygen uses p orbital to create π bond.

(50) **Answer :** (3)
Solution:
Resonating structures are not in equilibrium with each other.

(51) **Answer :** (2)
Solution:

Compounds	NF_3	NH_3	BF_3	CCl_4
Dipole moment (D)	0. 23	1. 47	0	0

(52) **Answer :** (3)
Hint:

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As the Bond order increases, the stability also increases.

Solution:

- $Li_2 \rightarrow Li_2^+$

BO 1 0.5

$(Diamagnetic$ $(Paramagnetic)$

- $O_2 \rightarrow O_2^+$

BO 2 2.5

$Paramagnetic$ $Paramagnetic$

- $N_2^+ \rightarrow N_2$

BO 2.5 3

$paramagnetic$ $Diamagnetic$

- $H_2 \rightarrow H_2^+$

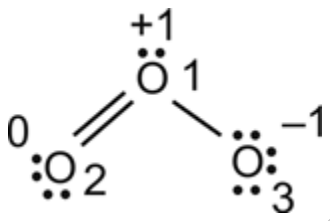
BO 1 0.5

$Diamagnetic$ $Paramagnetic$

(53) Answer : (3)

Solution:

Formal charge = $\left[V - L - \frac{1}{2}S\right]$



$\begin{bmatrix} V = \text{valence } e^- \\ L = \text{lone pair} \\ S = \text{Bonding } e^- \end{bmatrix}$

$\text{F.C on } O_1 = 6 - 2 - \frac{1}{2} \times 6 = +1$

$\text{F.C on } O_2 = 6 - 4 - \frac{1}{2} \times 2 = 0$

$\text{F.C on } O_3 = 6 - 6 - \frac{1}{2} \times 2 = -1$

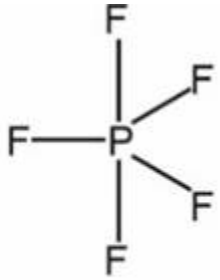
(54) Answer : (1)

Solution:

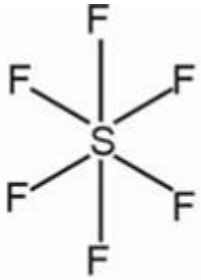
Bond type	Covalent bond length (pm)
C – H	107
C – N	143
C – C	154

(55) Answer : (2)

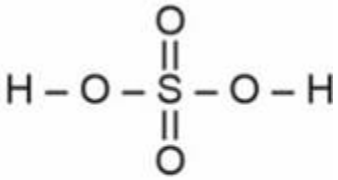
Solution:



10 electrons around P atom



12 electrons around S atom

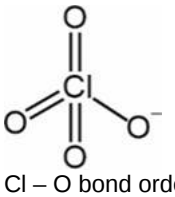


12 electrons around S atom

(56) Answer : (4)

Solution:

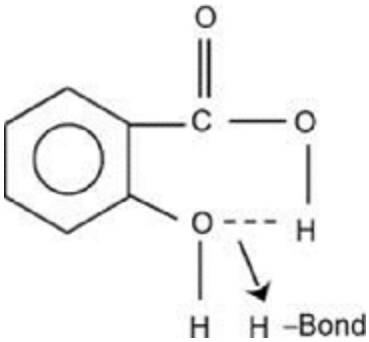
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Cl – O bond order = 7/4 = 1.75

(57) Answer : (2)
Solution:
C_2^{2-} ion contains 1 σ bond and 2π bonds.

(58) Answer : (3)
Solution:



(59) Answer : (4)
Solution:
NO is odd electron molecule.

(60) Answer : (4)
Solution:
For He_2 molecule

Electronic configuration is σ1 s^2, σ*1 s^2

so bond order = $\frac{1}{2}[N_b - N_a]$

$-\frac{1}{2}[2-2]$

$=0$

Since, bond order is zero, so He_2 molecule does not exist.

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(61) Answer : (1)
Hint:
Species having unpaired electrons are paramagnetic in nature.
Solution:
Electronic configuration of O_2 according to MOT

$O_2 : \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_z^2,$

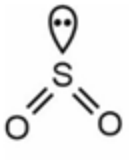
$\pi 2p_x^2 = \pi 2p_y^2, \pi^* 2p_x^1 = \pi^* 2p_y^1$

Since O_2 has two unpaired electron in π* molecular orbitals it will be paramagnetic in nature.

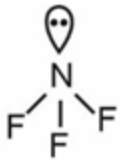
(62) Answer : (3)
Solution:



(T-shape)



(Bent)

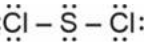


(Trigonal pyramidal)



(See-saw)

(63) **Answer :** (1)
Solution:



Octet theory is applicable to SCl_2 molecule.

(64) **Answer :** (4)
Solution:

% ionic character = $\frac{2.5\times3.33\times10^{-30}\times100}{1.6\times10^{-19}\times1.5\times10^{-10}}$ = 34.7%

(65) **Answer :** (4)
Solution:

Species	C–O Bond Order
CO_2	2
CO	3
CO_3^{2-}	1.33

(66) **Answer :** (4)
Solution:

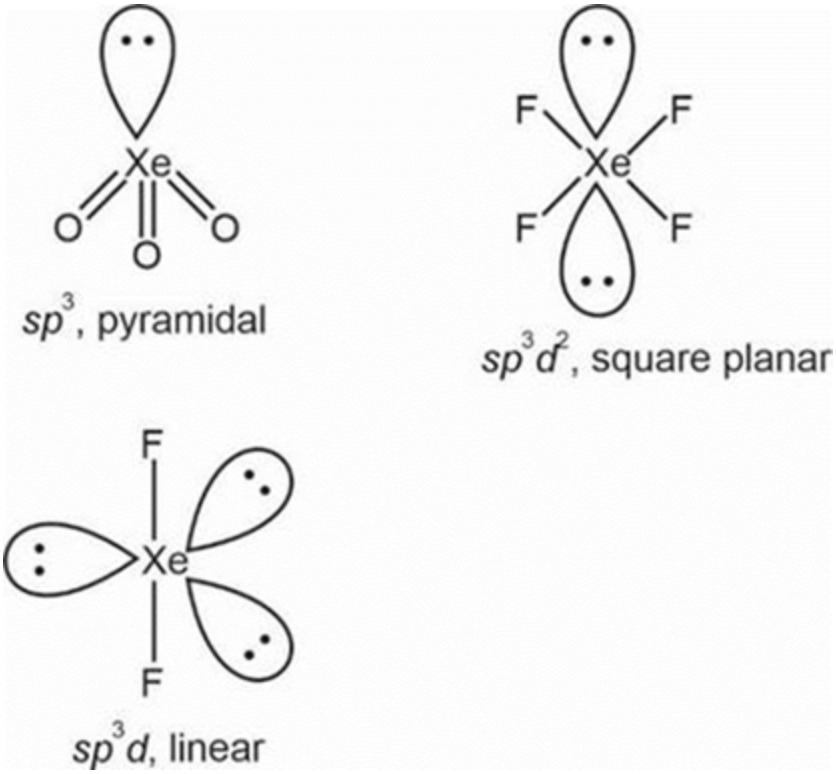
Species	Bond Order
O_2^-	1.5
O_2	2
O_2^+	2.5

Lesser is the bond order, longer is the bond length.

(67) **Answer :** (4)
Solution:
Correct order of lattice enthalpy
LiF > NaF > LiCl > NaCl

(68) **Answer :** (4)
Solution:

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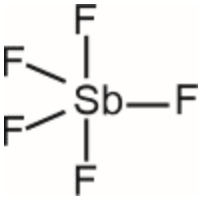


(69) **Answer :** (3)
Solution:
Boiling point of HF is higher than HCl due to intermolecular hydrogen bond.

(70) **Answer :** (2)
Solution:
MOT of N_2

$\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \begin{cases} \pi 2p_x^2 \\ \pi 2p_y^2 \end{cases}, \sigma 2p_z{}^2$

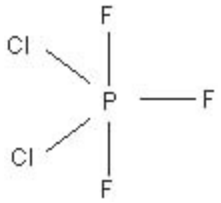
(71) **Answer :** (2)
Solution:
SbF_5 is trigonal bipyramidal hence there are maximum 4 elements in a plane.



(72) **Answer :** (2)
Solution:
O_2^{2-} is a diamagnetic species.
$O_2^{2-} = (\sigma 1s)^2 (\sigma^* 1s)^2 (\sigma 2s)^2 (\sigma^* 2s)^2 (\sigma 2p_z)^2$
$(\pi 2p_x)^2 = (\pi 2p_y)^2 (\pi^* 2p_x)^2 = (\pi^* 2p_y)^2$

(73) **Answer :** (1)
Solution:

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$\mu \neq 0$ ∴(polar molecule)

(74) **Answer :** (2)
Solution:
If



has x dipole moment, (i) and (iii) has x dipole moment, (ii) has 2x dipole moment, (iv) and (v) has zero dipole moment.

(75) **Answer :** (1)
Solution:
In $N(SiH_3)_3$ p π – d π back bonds are formed.

(76) **Answer :** (3)
Solution:
Promotion of electron is not essential condition prior to hybridisation.

(77) **Answer :** (3)
Hint:
The greater the charge on the cation, the greater the covalent character of the ionic bond.
Solution:
The polarising power of cation increases with increase in charge, hence covalent character of ionic bond increases.
So increasing order of covalent character is
$NaCl < MgCl_2 < AlCl_3$

(78) **Answer :** (3)
Solution:
According to VSEPR theory repulsive interaction decrease in the order lp – lp > lp – bp > bp – bp

(79) **Answer :** (4)
Solution:

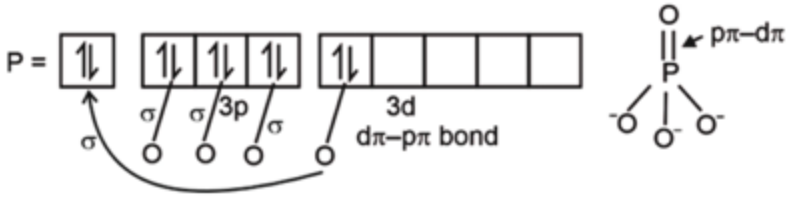
M.O. configuration of NO^+ is
$\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \pi 2p_x{}^2 = \pi 2p_y{}^2, \sigma 2p_z{}^2$

(80) **Answer :** (4)
Solution:
XeF_4 has total 14 lone pair of electrons.

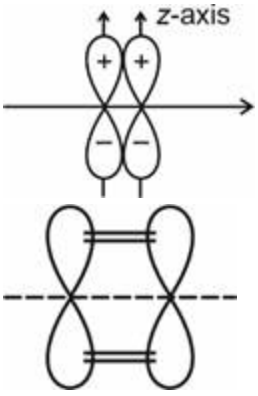
(81) **Answer :** (3)
Hint:
Electrons are filled in the orbitals on the basis of increasing energy level.
Solution:
$F_2 = \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_z^2$
$\pi 2p_x^2 = \pi 2p_y^2 \ \pi^* 2p_x^2 \ = \pi^* 2p_y^2$
$N_2 = \sigma 1s^2 \ \sigma^* 1s^2 \ \sigma 2s^2 \ \sigma^* 2s^2 \ \pi 2p_x^2 = \ \pi p_y^2 \ \sigma 2p_z^2$

(82) **Answer :** (2)
Solution:
*d*π – *p*π bond *i.e.* PO_4^{3-} because P have d-orbital

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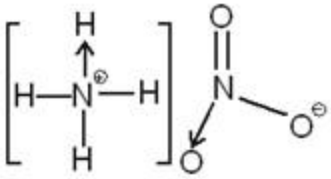


(83) **Answer :** (3)
Hint:
π bond is formed by sideways overlapping.
Solution:



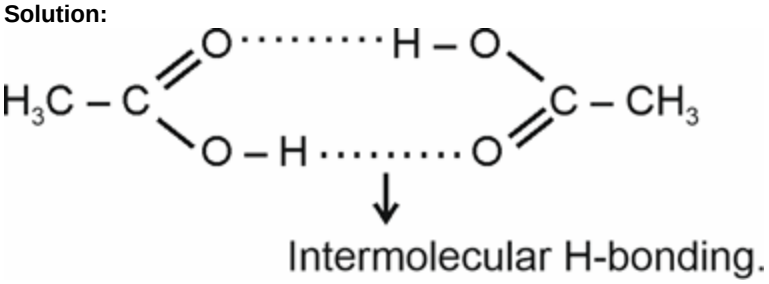
The atomic orbitals are overlapping sideways, so, π bond will be formed.

(84) **Answer :** (2)
Solution:

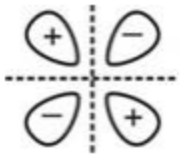


(85) **Answer :** (1)
Solution:
An atomic orbital is monocentric as it is influenced by one nucleus only and molecular orbital is polycentric as it is being influenced by 2 or more nuclei.

(86) **Answer :** (2)
Solution:



(87) **Answer :** (2)
Solution:



represents π-antibonding molecular orbital.

(88) **Answer :** (3)

Solution:
Number of hybrid orbitals are equal to the number of atomic orbitals involved in hybridisation.

(89) Answer : (2)
Solution:
In CCl_4, the number of 109.5° bond angles is six.

(90) Answer : (4)
Solution:
PH_3 is pyramidal
XeO_3 is pyramidal

BOTANY

(91) Answer : (3)
Solution:
Archaebacteria have different cell wall composition as compared to eubacteria which have peptidoglycan in their cell wall.

(92) Answer : (4)
Solution:
Cellular organisation of the body is the defining feature of life forms. Viruses are not considered to be a cell.

(93) Answer : (1)
Solution:
In members of Deuteromycetes, sexual reproduction is absent and therefore, they are called fungi imperfecti.

(94) Answer : (3)
Solution:
Bacteria are mainly classified under four categories on the basis of their shapes. Some of the bacteria play a great role in recycling nutrients. *Mycoplasma* lack cell wall.

(95) Answer : (4)
Solution:
Plasmodium is considered to be the most notorious protozoan causing malaria.

(96) Answer : (3)
Solution:
Fusion of two somatic cells $\rightarrow$Formation of dikaryotic structure that gives rise to basidium $\rightarrow$Karyogamy within basidium $\rightarrow$Spores formed on surface of basidium $\rightarrow$Maturation and rupturing of basidiocarp.

(97) Answer : (4)
Solution:
Fungi is with cell wall. Therefore these were placed in kingdom Plantae by Linnaeus.

(98) Answer : (2)
Solution:
a - Virus, b - Viroid, c - Bacteriophage, d - Lichen

(99) Answer : (2)
Hint:
The given figure is of *Aspergillus.*
Solution:
Aspergillus belong to the class Ascomycetes (sac-fungi). It produces conidia exogenously.

(100) Answer : (1)
Solution:
Bacterial structure is very simple but it has complex behaviour

(101) Answer : (3)
Solution:
Agaricus (Mushrooms) forms basidiocarps which are edible.

(102) Answer : (2)
Solution:
A = vegetative cell
B = Heterocyst
C = Mucilagenous sheath

(103) Answer : (3)
Hint:
Saccharomyces (yeast) is a unicellular fungus.
Solution:
Yeast does not produce fruiting body while *Agaricus,* morels and truffles produce edible fruiting bodies.

(104) Answer : (3)
Solution:
The parasitic forms of flagellated protozoans cause diseases such as sleeping sickness. Example: *Trypanosoma.*

(105) Answer : (3)
Solution:
Cuscuta is a parasitic plant. *Gonyaulax* causes red tides in sea.

(106) Answer : (3)
Solution:
Membrane bound cell organelles are found in eukaryotes.

(107) Answer : (2)
Solution:
Sexual reproduction is absent in the members of Deuteromycetes.
Colletotrichum, *Alternaria* and *Trichoderma* are the members of Deuteromycetes.

(108) Answer : (4)
Solution:
Diatoms are the chief producers in the oceans.

(109) Answer : (2)
Solution:
Prions cause bovine spongiform encephalopathy commonly called mad cow disease in cattle.

(110) Answer : (3)
Solution:
Methanogens are methane producing bacteria abundant in marshy area and gut of ruminants.

(111) Answer : (2)
Solution:
• *Nostoc* and *Anabaena* are cyanobacteria.
• Fungi has mainly chitinous cell wall except in oomycetes

(112) Answer : (1)
Solution:
The cell wall of diatoms is embedded with silica.

(113) Answer : (1)
Solution:
Mycorrhiza is a symbiotic association between roots of higher plants and fungi.

(114) Answer : (2)
Solution:
Rod shaped - *Bacillus*, spherical-*Coccus*, spiral-*Spirillum*, comma shaped - *Vibrium*.

(115) Answer : (1)
Solution:
Bacteriophages or bacterial viruses usually have dsDNA as their genetic material.

(116) Answer : (2)
Solution:
Euglenoids show mixotrophic mode of nutrition.

(117) Answer : (4)
Hint:
Spirogyra is not a unicellular organism.
Solution:
Kingdom Protista has brought together *Chlamydomonas, Chlorella* with *Paramoecium* and *Amoeba*.

(118) Answer : (3)
Solution:
Albugo causes white rust in mustard.

(119) Answer : (3)

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Solution:
Wheat rust is caused by *Puccinia graminis*.

(120) Answer : (1)
Solution:
Biological classification of plants and animals was first proposed by Aristotle on the basis of simple morphological characters.

(121) Answer : (2)
Solution:
Prions are similar in size to viruses.

(122) Answer : (2)
Solution:
Zygospores are diploid sexual spores formed by members of Phycomycetes.

(123) Answer : (1)
Solution:
Ascospores are haploid, endogenous meiospores.

(124) Answer : (3)
Hint:
In Monera, cyanobacteria show oxygenic photosynthesis.
Solution:
In five kingdom classification, chlorophyllous photosynthetic organisms evolving oxygen belong to three kingdoms (Monera, Protista and Plantae).

(125) Answer : (2)
Hint:
The kingdom, whose sole members are bacteria is divided into two different domains in six kingdom classification.
Solution:
In three domain system, kingdom Monera is divided into two different domains.

(126) Answer : (1)
Hint:
Heterotrophic bacteria are most abundant in nature.
Solution:
Chemosynthetic autotrophic bacteria oxidise various inorganic substances such as nitrates, nitrites and ammonia and use the released energy for their ATP production. Heterotrophic bacteria are important decomposers.

(127) Answer : (4)
Solution:
Heterotrophic bacteria are helpful in making curd from milk.
Chemoautotrophic bacteria help in recycling of nutrients.

(128) Answer : (2)
Solution:
Slime moulds form the plasmodium under the suitable conditions. During unfavourable conditions, the plasmodium differentiates and forms fruiting bodies bearing spores.

(129) Answer : (4)
Hint:
Except sporozoans, all protozoan groups have their specific locomotory structures.
Solution:
They all are single celled and heterotrophs.
Only ciliated protozoans have specific regions for ingestion and egestion i.e. gullet and cytopyge

(130) Answer : (3)
Solution:
Euglenoids have heterokont flagellation, *i.e.*, one long and one short flagella.

(131) Answer : (3)
Solution:
Members of fungi imperfecti have branched and septate mycelium. They form conidia and help in mineral recycling.

(132) Answer : (1)
Solution:
Infectious nature of viruses is decided by its nucleic acid. Virus is a nucleoprotein in which genetic material is infectious in nature.

(133) Answer : (3)

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Solution:
In members of Phycomycetes, asexual reproduction indeed takes place by zoospores or by aplanospores. Phycomycetes are known for their asexual reproduction through these spores.
Members of Ascomycetes can be saprophytic, decomposers, parasitic, or coprophilous.

(134) Answer : (3)
Hint:
In members of Ascomycetes and Basidiomycetes, an intervening dikaryotic stage occurs.
Solution:
Agaricus and *Aspergillus* belong to Basidiomycetes and Ascomycetes respectively.

(135) Answer : (2)
Solution:
Prions cause mad cow disease.

ZOOLOGY

(136) Answer : (2)
Solution:
Receptors associated with aortic arch and carotid artery can recognise changes in CO_2 and H^+ concentration and send necessary signals to the rhythm centre for remedial actions. The role of oxygen in the regulation of respiratory rhythm is quite insignificant.

(137) Answer : (3)
Hint:
Protein that is the main content of bird's egg.
Solution:
Albumin helps in maintaining the blood colloidal osmotic pressure.
- Fibrinogen and prothrombin are blood clotting proteins.
- Globulins are involved in defence mechanisms of the body.

(138) Answer : (4)
Solution:
Plasma – Clotting factors = Serum

(139) Answer : (3)
Solution:
Receptors associated with aortic arch and carotid artery can recognise changes in CO_2 and H^+ concentration and send necessary signals to the rhythm centre for remedial actions. The role of oxygen in the regulation of respiratory rhythm is quite insignificant.

(140) Answer : (4)
Solution:
Blood becomes alkaline when there is high pO_2 and less pCO_2.

(141) Answer : (2)
Solution:
Formed elements constitute about 45 per cent of the total blood composition.

(142) Answer : (4)
Solution:
Each haemoglobin molecule can carry a maximum of four molecules of O_2.

(143) Answer : (2)
Hint:
Phylogenetically closest to birds
Solution:
Reptiles have 3-chambered heart, except crocodiles, which possess 4-chambered heart with two atria and two ventricles.

(144) Answer : (3)
Hint:
Heart failure is sometimes referred to as congestive heart failure.
Solution:
Heart failure:- Heart does not pump blood effectively enough to meet the needs of the body.

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Cardiac arrest:- When the heart stops beating.

Heart attack:- When the heart muscle is suddenly damaged by inadequate blood supply.

(145) Answer : (4)

Hint:

ST segment is elevated in myocardial infarction.

Solution:

Enlargement of P-wave indicates enlargement of atria.
Enlarged Q and R waves indicate myocardial infarction (Heart attack).
The S-T segment is elevated in myocardial infarction and depressed when the heart muscles receive insufficient oxygen.
T-wave is flat when heart muscle receives insufficient oxygen.

(146) Answer : (1)

Solution:

Nearly, 20-25 per cent of CO_2 is transported by RBCs as carbamino-haemoglobin whereas, 70 per cent of it is carried as bicarbonate. About 7 per cent of CO_2 is carried in a dissolved state through plasma. CO binds to Hb and forms carboxy-haemoglobin.

(147) Answer : (2)

Solution:

At the alveolar site where pCO_2 is low, the reaction proceeds in the opposite direction leading to the formation of CO_2 and H_2O. Thus, CO_2 trapped as bicarbonate at the tissue level and transported to the alveoli is released out as CO_2.

(148) Answer : (1)

Hint:

Maximum O_2 is transported as oxyhaemoglobin.

Solution:

Mode of transport	**O_2(%)**	**CO_2(%)**
RBC	97	20-25
Plasma	3	7

(149) Answer : (2)

Hint:

Leads to the closure of tricuspid valve.

Solution:

The QRS complex represents the depolarisation of the ventricles which initiates the ventricular contraction.
P-wave represents the electrical excitation (depolarisation) of the atria.
T-wave represents the return of the ventricles from excited to the normal state (repolarisation).

(150) Answer : (1)

Hint:

Type of leucocytes present in least amount.

Solution:

Basophils – 0.5-1 per cent of total WBCs
Eosinophils – 2-3 per cent of total WBCs
Neutrophils – 60-65 per cent of total WBCs
Monocytes – 6-8 per cent of total WBCs
Lymphocytes – 20-25 per cent of total WBCs

(151) Answer : (1)

Solution:

Exchange of gases between lungs and blood takes place by simple diffusion only.

(152) Answer : (2)

Solution:

A sigmoid curve is obtained when percentage saturation of haemoglobin with oxygen is plotted against the partial pressure of oxygen (Oxyhemoglobin dissociation curve).

(153) Answer : (2)

Solution:

Various substances dissolved in plasma are transported to different tissues of the body.

(154) Answer : (2)

Hint:

AVN is the pacesetter of heart.

Solution:

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SAN has the highest inherent capacity of impulse generation i.e. it generates highest, number of action potentials per minute.

(155) Answer : (3)
Solution:
Left auricle and left ventricle in higher vertebrates including humans is considered as systemic heart whereas right auricle and right ventricle which pumps blood to lungs is called pulmonary heart. Mitral valve/bicuspid valve is present between left auricle and left ventricle of the heart.

(156) Answer : (4)
Solution:
Under normal physiological conditions, 5 mL of O_2 is transported to tissues by 100 mL of oxygenated blood.

(157) Answer : (1)
Solution:
Asbestosis results from inhalation of asbestos particles which can result in severe fibrosis and high risk for cancer.

(158) Answer : (2)
Hint:
Each ventricle pumps about 70 ml blood per beat.
Solution:
Heart beats about 72 times per minute so cardiac output of each ventricle is about 72 × 70 = 5040 mL, so both the ventricles pump about 10 L of blood in one minute.

(159) Answer : (3)
Hint:
Thrombocytes are platelets.
Solution:
Leukocytes averages 6000 to 8000 mm^{-3} of blood.

(160) Answer : (3)
Hint:
Universal solvent
Solution:
90 – 92 per cent of plasma is water and proteins contribute 6 – 8 per cent of it. Fibrinogens, globulins and albumins are the major proteins present in the blood plasma.

(161) Answer : (1)
Solution:
Since, the human heart is autoregulated by the patches of nodal tissues (SA node, AV node) which are made up of cardiac muscles, it is said to be myogenic in nature.

(162) Answer : (2)
Solution:
Systemic arteries have oxygenated blood having pO_2 95 mm Hg and pCO_2 40 mm Hg.

Respiratory Gas	Atmospheric Air	Alveoli	Blood (Deoxygenated)	Blood (Oxygenated)	Tissues
O_2	159	104	40	95	40
CO_2	0.3	40	45	40	45

(163) Answer : (1)
Solution:
CO_2 is carried by haemoglobin as carbamino-haemoglobin (about 20-25 percent).

(164) Answer : (3)
Solution:
Medulla oblongata moderates the cardiac activity through autonomic nervous system.

(165) Answer : (2)
Solution:
In acidic conditions, affinity of haemoglobin for oxygen decreases and oxyhemoglobin dissociation curve shifts to right side.

(166) Answer : (2)
Solution:
Blood moves out of the ventricles into the respective vessels during ventricular systole and atrio-ventricular valves are closed during ventricular systole.

(167) Answer : (3)

Hint:
These are also called platelets.
Solution:
Thrombocytes are cell fragments produced from megakaryocytes. Thrombocytes/platelets release a variety of substances involved in coagulation of blood.

(168) Answer : (3)
Solution:
Fishes have 2-chambered heart with an atrium and a ventricle. Amphibians and reptiles (except crocodile) have 3-chambered heart with 2 atria and a single ventricle, whereas, birds and mammals possess 4-chambered heart with two atria and two ventricles.

(169) Answer : (2)
Hint:
Portal circulation is present between liver and intestine.
Solution:
Blood from stomach, small intestine and large intestine is carried to liver through hepatic portal vein for nutrient processing and detoxification.

(170) Answer : (3)
Solution:
Emphysema is a chronic disorder in which alveolar walls are damaged due to which respiratory surface is decreased. Major causes are cigarette smoking and the inhalation of other smoke or toxic substances over a period of time.

(171) Answer : (2)
Solution:
In total, blood flows 72 times per minute both in pulmonary and systemic circulation in a normal resting human.

(172) Answer : (2)
Solution:
Neutrophils and monocytes are phagocytic cells which destroy foreign organisms entering the body.

(173) Answer : (3)
Solution:
Arthropods like cockroach, silkworm (*Bombyx*) and butterfly have open circulatory system.
Earthworm is an annelid and exhibits closed circulatory system.

(174) Answer : (1)
Hint:
Closure of tricuspid and bicuspid valves produces first heart sound.
Solution:
During each cardiac cycle, two prominent sounds are produced which can be easily heard through a stethoscope. The first heart sound (lub) is associated with the closure of the tricuspid and bicuspid valves whereas the second heart sound (dub) is associated with the closure of the semilunar valves. These sounds are of clinical diagnostic significance.

(175) Answer : (2)
Solution:
Factors favour the formation of oxyhaemoglobin in alveoli are :
- High pO_2
- Low pCO_2
- Lesser H^+ concentration
- Lower temperature

(176) Answer : (2)
Hint:
In alveoli, pO_2 is 104 mmHg and pCO_2 is 40 mmHg.
Solution:

1. In alveoli of lungs, pO_2 is high but pCO_2 is low in comparison to venous blood. Respiratory rhythm centre regulates normal respiratory rate and is found in medulla oblongata.

(177) Answer : (3)
Hint:
Mother is Rh negative in this condition.
Solution:
Injecting Rh antigen to mother's blood leads to agglutination reactions. In this condition, anti-Rh antibodies are administered to the mother immediately after the delivery of the first child.

(178) Answer : (4)

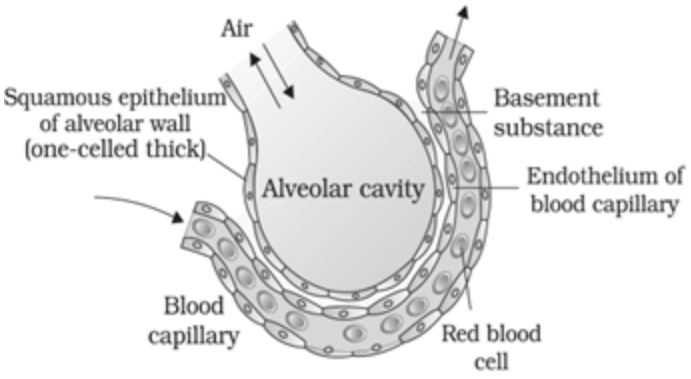
Solution:

Individuals with blood group AB lack antibodies in their blood plasma.

(179) Answer : (2)

Solution:

Endothelium of capillaries + Basement substance + Simple squamous epithelium of alveoli together constitute respiratary membrane.



(180) Answer : (1)

Hint:

The product of one step acts as a catalyst for the next step in a cascade process.

Solution:

Prothrombin is converted to thrombin by thrombokinase which, in turn, converts fibrinogens to fibrins.

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